Semi-supervised learning

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Lecture plan

1. Semi-Supervised Learning (SSL)
2. Mechanisms of missing labels: MCAR, MAR, MNAR
3. Two popular methods:
    - Self-Training Method
    - Label Propagation Method

Semi-Supervised Learning

Binary classification under semi-supervised learning:

- Response/target variable $Y \in \mathcal{Y}$, where $\mathcal{Y} = \{0, 1\}$ (e.g. presence of disease)
- Feature/attribute/explanatory variable vector $X = (X_1, \ldots, X_p) \in \mathcal{X}$, where $\mathcal{X}$ denotes the feature space (e.g. clinical variables)
- For some observations, labels are not assigned (e.g. no diagnosis available, we do not know whether the patient is healthy or sick)

Training data:

X_1	X_2	...	X_p	Y
1.0	2.2	...	4.2	1
2.4	1.3	...	3.1	0
0.9	1.4	...	3.2	1
0.6	1.2	...	3.2	0
1.2	3.5	...	7.2	?
1.7	3.2	...	3.2	?
...	...	...	...	...
1.7	3.2	...	3.2	**?**

Question: can unlabeled observations help us fit the model? Question: can we infer labels for observations without assigned labels?

Semi-Supervised Learning

Semi-supervised learning methods are useful in situations where assigning (annotating) labels is difficult and costly, while obtaining observations and feature values is relatively easy.

Example applications:

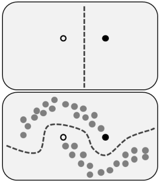
1. Speech recognition, e.g. recognizing whether a voice belongs to a male or female speaker; whether the voice indicates positive or negative emotions; whether the voice indicates the presence of disease (stroke, mental disorders).
2. Detection of illegal content (texts, images) in social networks. Large amounts of unlabeled content.
3. Sentiment analysis, e.g. recognizing whether an article has a positive or negative sentiment. Large amounts of unlabeled text.
4. Survey studies. Some respondents do not answer certain questions (e.g. about income or political views).

Semi-Supervised Learning

- Labeled data: $L = \{(x_1, y_1), \dots, (x_l, y_l)\}$, where $y_1, , \dots, y_l$ are observable.
- Unlabeled data: $U = \{(x_{l+1}, y_{l+1}), \dots, (x_{l+u}, y_{l+u})\}$, where $y_{l+1}, , \dots, y_{l+u}$ are **NOT** observable.
- Feature vector $x_i \in \mathcal{X}$, e.g. $\mathcal{X} = R^p$.
- Very often $l << u$.

Naive approach: build the model only using labeled data $L = \{(x_1, y_1), \dots, (x_l, y_l)\}$.

This approach may be insufficient when l is small (e.g. in extreme cases we may only have a few or a dozen observations).



Mechanisms of missing data

We consider variables (X, Y), where:

- X – observed features,
- Y – variable that may be unobserved.

We introduce an indicator variable:

$$S = \begin{cases} 1 & \text{value observed} \\ 0 & \text{value missing} \end{cases}$$

The missingness mechanism describes the probability:

$$P(S = 1 \mid X, Y)$$

We distinguish three basic mechanisms:

1. MCAR – Missing Completely At Random,
2. MAR – Missing At Random,
3. MNAR – Missing Not At Random.

MCAR – Missing Completely At Random

Definition:

$$P(S = 1 \mid X, Y) = P(S = 1)$$

Probability of missingness:

- does not depend on features X,
- does not depend on the value of Y.

For the MCAR mechanism we have:

$$P(X, Y|S = 1) = \frac{P(S = 1|X, Y)P(X, Y)}{P(S = 1)} = P(X, Y)$$

Labeled data are representative of the entire population.

MCAR – Missing Completely At Random

Definition:

$$P(S = 1 \mid X, Y) = P(S = 1)$$

Probability of missingness:

- does not depend on features X,
- does not depend on the value of Y.

Risk function:

$$R(h) = E_{P(X,Y)} l(Y, h(X)) = E_{P(X,Y|S=1)} l(Y, h(X))$$

Conclusion: minimizing risk on labeled data corresponds to minimizing population risk.

MAR – Missing At Random

Definition:

$$P(S = 1 \mid X, Y) = P(S = 1|X)$$

Probability of missingness:

- depends on features X,
- does not depend on the value of Y after conditioning on X.
- Example: X- age, Y-income. Younger individuals skip the income question more often.

For the MAR mechanism we have:

$$P(X, Y|S = 1) = \frac{P(S = 1|X, Y)P(X, Y)}{P(S = 1)} = \frac{P(S = 1|X)}{P(S = 1)} P(X, Y)$$

Labeled data are NOT representative of the entire population.

MAR – Missing At Random

Definition:

$$P(S = 1 \mid X, Y) = P(S = 1|X)$$

Probability of missingness:

- depends on features X,
- does not depend on the value of Y after conditioning on X.
- Example: X- age, Y-income. Younger individuals skip the income question more often.

For the MAR mechanism we have:

$$P(Y|X, S = 1) = \frac{P(S = 1|Y, X)P(Y|X)}{P(S = 1|X)} = P(Y|X)$$

Conclusion: the posterior probability does not change.

MAR – Missing At Random

For the MAR mechanism we have:

$$P(X,Y|S=1)=\frac{P(S=1|X,Y)P(X,Y)}{P(S=1)}=\frac{P(S=1|X)}{P(S=1)}P(X,Y)$$

Inverse Probability Weighting (IPW):

Risk function:

$$R(h)=E_{P(X,Y)}l(Y,h(X))=E_{P(X,Y|S=1)}\left(\frac{P(X,Y)}{P(X,Y|S=1)}l(Y,h(X))\right)=$$

$$E_{P(X,Y|S=1)}\left(\underbrace{\frac{P(S=1)}{P(S=1|X)}}_{=w(X)}l(Y,h(X))\right)$$

Conclusion: correction of the risk function is required by introducing weights

$$w(x)=\frac{P(S=1)}{P(S=1|X=x)}$$

MAR – Missing At Random

Risk function:

$$R(h) = E_{P(X,Y)} l(Y, h(X)) = E_{P(X,Y|S=1)} \left(\frac{P(X,Y)}{P(X,Y|S=1)} l(Y, h(X)) \right) =$$

$$E_{P(X,Y|S=1)} \left(\underbrace{\frac{P(S=1)}{P(S=1|X)}}_{=w(X)} l(Y, h(X)) \right)$$

Conclusion: correction of the risk function is required by introducing weights

$$w(x) = \frac{P(S=1)}{P(S=1|X=x)}$$

Weights $w(x)$ can be estimated. It is sufficient to estimate $P(S=1)$ (fraction of labeled observations) and $P(S=1|X)$ (e.g. using logistic regression).

MNAR – Missing Not At Random

Probability of missingness:

$$P(S = 1 \mid X, Y)$$

- depends on features X and Y
- Example: X - age, Y - income. Younger individuals with lower income are more likely to skip the income question.

For the MNAR mechanism we have:

$$P(X, Y|S = 1) = \frac{P(S = 1|X, Y)}{P(S = 1)} P(X, Y)$$

Labeled data are NOT representative of the entire population.

MNAR – Missing Not At Random

Probability of missingness:

$$P(S = 1 \mid X, Y)$$

- depends on features X and Y
- Example: X - age, Y - income. Younger individuals with lower income are more likely to skip the income question.

For the MNAR mechanism we have:

$$P(Y|X, S = 1) = \frac{P(S = 1|Y, X)P(Y|X)}{P(S = 1|X)} \neq P(Y|X)$$

Conclusion: the posterior probability changes.

MNAR – Missing Not At Random

For the MNAR mechanism we have:

$$P(X,Y|S=1) = \frac{P(S=1|X,Y)}{P(S=1)} P(X,Y)$$

Inverse Probability Weighting (IPW):

Risk function:

$$R(h) = E_{P(X,Y)} l(Y,h(X)) = E_{P(X,Y|S=1)} \left(\frac{P(X,Y)}{P(X,Y|S=1)} l(Y,h(X)) \right) =$$

$$E_{P(X,Y|S=1)} \left(\underbrace{\frac{P(S=1)}{P(S=1|X,Y)}}_{=w(X,Y)} l(Y,h(X)) \right)$$

Conclusion: correction of the risk function is required by introducing weights (Y not observed for $S=0$)

$$w(x,y) = \frac{P(S=1)}{P(S=1|X=x, Y=y)}$$

Summary

We distinguish three basic mechanisms:

1. MCAR – Missing Completely At Random,
2. MAR – Missing At Random,
3. MNAR – Missing Not At Random.

Remarks:

- Most existing SSL (Semi-Supervised Learning) methods assume **MCAR** or **MAR**, although this is very rarely stated explicitly.
- Even in the MCAR setting, SSL methods may improve the performance of the naive method, despite the fact that it is asymptotically unbiased (the theoretical risk of the naive method equals the population risk).
- The MNAR mechanism is very challenging because we cannot estimate the weights and therefore cannot correct the risk function.

Self-Training Method [1]

Algorithm:

- Initialization of pseudo-labels: $\tilde{y}_i = y_i$ for $i \in L$ and $\tilde{y}_i = NA$ for $i \in U$.
- Repeat the following steps:
  1. Fit a model $h : \mathcal{X} \to [0,1]$ using data L.
  2. Perform prediction $h(x_i) = \hat{P}(y_i = 1|x_i)$, for $i \in U$.
  3. If:
     - $h(x_i) > \tau$ then $\tilde{y}_i = 1$, for $i \in U$.
     - $h(x_i) < 1 - \tau$ then $\tilde{y}_i = 0$, for $i \in U$.
  4. Let $A = \{i \in U : \tilde{y}_i \neq NA\}$.
  5. $L \leftarrow L \cup A$.

Parameter τ is a threshold; commonly used values are $\tau = 0.95, \tau = 0.9$, $\tau = 0.75$ (default value in scikit learn).

Another possibility: at each step we assign pseudo-labels to k observations from set U that have the highest posterior probability for either class (1 or 0); $k = 10$ is the default value in scikit learn.

[1] We consider the binary classification case

Label Propagation Method [2]

- Main idea of the algorithm: neighboring points should have the same labels. Labeled observations 'emit' labels to nearby unlabeled observations: 'labeled data act as sources that push out labels through unlabaled data. The algorithm propagates labels through dense unlabeled data regions.'
- We construct a graph in which vertices correspond to observations: $x_1, \ldots, x_l, x_{l+1}, \ldots, x_{l+u}$ (both from set L and set U).
- Each edge between vertices i and j is assigned a weight; the closer the observations are (in terms of Euclidean distance), the larger the weight:

$$w_{ij} := \exp\left(-\frac{d_{ij}^2}{\sigma}\right) = \exp\left(-\frac{\sum_{k=1}^{p}(x_i^k - x_j^k)^2}{\sigma^2}\right),$$

  where σ is a parameter and $x_i = (x_i^1, \ldots, x_i^p)$.
- Other choices of distance measures are also possible, depending on the type of variables considered (binary, discrete).

[2] https://pages.cs.wisc.edu/~jerryzhu/pub/CMU-CALD-02-107.pdf

Label Propagation Method [3]

- We define a $(l+u) \times (l+u)$ transition probability matrix:

$$T_{ij} := P(j \to i) = \frac{w_{ij}}{\sum_{k=1}^{l+u} w_{kj}}$$

- Large values of T_{ij} mean that the label assigned to observation j will be easily propagated to observation i.
- Additionally, we define a matrix Y of dimension $(l+u) \times C$, where C denotes the number of classes, such that the i-th row of matrix Y describes the class distribution for the i-th observation.

  For example: $Y[i,:] = (0,1,0,0,0)$ means that the observation belongs to class 2.

[3] https://pages.cs.wisc.edu/~jerryzhu/pub/CMU-CALD-02-107.pdf

Label Propagation Method [4]

Algorithm:

- Initialize Y with random values for rows corresponding to observations from set U.
- Repeat the following steps:
  1. Label propagation:
  $$Y \leftarrow TY$$
  2. Normalize rows of matrix Y.
  3. For labeled observations $i \in L$:
  $$Y_{ic} = \delta(y_i, c),$$
  where $\delta(y_i, c) = 1$ if $y_i = c$ and $\delta(y_i, c) = 0$ otherwise.

[4] https://pages.cs.wisc.edu/~jerryzhu/pub/CMU-CALD-02-107.pdf

Label Propagation Method

Example:

- We have 4 observations: x_1, x_2, x_3, x_4, such that $y_1 = 0$, $y_2 = 1$, $y_3 = ?$, $y_4 = ?$
- Suppose that $d_{13} \approx 0$, $d_{24} \approx 0$.
- Therefore, T_{13} and T_{24} take large values, while T_{12}, T_{14}, T_{23} and T_{34} take small values.
- Thus:

$$T = \begin{bmatrix} T_{11} & T_{12} & T_{13} & T_{14} \\ T_{21} & T_{22} & T_{23} & T_{24} \\ T_{31} & T_{32} & T_{33} & T_{34} \\ T_{41} & T_{42} & T_{43} & T_{44} \end{bmatrix} \approx \begin{bmatrix} T_{11} & 0 & T_{13} & 0 \\ 0 & T_{22} & 0 & T_{24} \\ T_{31} & 0 & T_{33} & 0 \\ 0 & T_{42} & 0 & T_{44} \end{bmatrix}$$

- In the first step:

$$Y = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$$

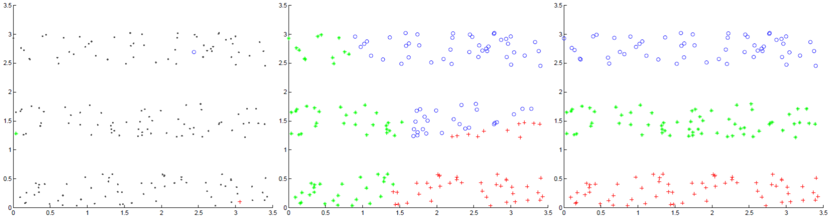
- After assigning $Y \leftarrow TY$ we obtain:

$$Y_{31} = T_{31} + 0.5T_{33} \qquad Y_{32} = 0.5T_{33}$$

- Since $Y_{31} > Y_{32}$, we assign observation x_3 to class 0.

Label Propagation Method

Example of the Label Propagation algorithm:

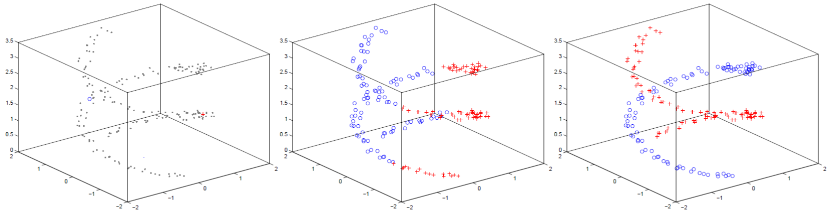


(a) The data (b) kNN (c) Label propagation

source: `https://pages.cs.wisc.edu/~jerryzhu/pub/CMU-CALD-02-107.pdf`

Label Propagation Method

Example of the Label Propagation algorithm:



(a) The data (b) kNN (c) Label propagation

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